

College of Engineering Educational Policy Committee Meeting Minutes: 2015

Minutes of the Tuesday September 1, 2015 College of Engineering Educational Policy Committee Meeting

Location: Room 115 SEO

In Attendance: Vahe Caliskan, Houshang Darabi, John Hetling, Joe Hummel, Eduard Karpov, Ludwig Nitsche, Alan Zdunek

1. Minutes from last meeting were not available; Ludwig Nitsche will locate minutes for approval.
2. Joe Hummel and Alan Zdunek were elected Chair and Secretary of the committee, respectively.
3. IBHE update: Directors of Graduate Studies are leading the effort in each department. Committee members will contact their DGS about what is needed from EPC or from the DUGS
4. Preparation for upcoming Administration Council meeting to be held Thursday, September 10, 10am-noon. Meeting will focus entirely on undergraduate issues.
 - a. Discussed strategic planning and what admission criteria are relevant (test scores, HS GPA, etc), what statistics are relevant, what they mean and do we need other means of assessing our student population.
 - b. FESP committee's ENGR100 recommendation will be discussed at the Admin Council meeting and was briefly discussed on how to get buy-in from the departments. What does 'interdisciplinary' truly mean in the context of engineering education?
 - c. DUGS to think about above issues for further discussion at Admin Council meeting.
5. Large classroom requests for Spring 2015 were discussed; To be addressed again in October/November.
6. COE program changes responding to LAS credit hour changes. A possible solution was proposed; use an engineering math course such as ECE 341, IE 342, IE 473 or a statistics course to make up the technical elective credit. It is not known whether this would be ABET accepted.
7. CHE Prerequisite change in CHE 311 to include MATH 220 was approved.
8. BioE, CS, and CME have items in the pipeline for EPC;
 - a. BioE working with CS to create new BS in Bioinformatics program, will be submitting a program revision to be effective Fall 2016, plus may have new courses to approve plus clearing out of 'defunct' courses

Respectfully submitted;
Alan Zdunek, Secretary

Minutes of the Tuesday September 8, 2015
College of Engineering Educational Policy Committee Meeting

Location 115 SEO

In Attendance: Vahe Caliskan, David Eddington, Eduard Karpov, Miiri Kotche, Ludwig Nitsche, Renata Revelo, Alan Zdunek

1. Minutes from September 1, 2015 meeting were approved
2. Course Request Change from Eduard Karpov to update prerequisites to CME 427 Engineering Hydrology were approved.
3. Discussion in preparation for September 10th Administration Council meeting focusing on undergraduate engineering education:
 - a. Rethinking our engineering programs- Do engineering students need an 'identity', a sense of 'belonging' to the engineering discipline.
 - b. Engineering student --> Engineering professional: How do we accomplish this transition at UIC. Do we work backwards from engineering applications/professions, incorporate more practical examples of using theory/math/physics, etc.
 - c. What about having upper-class mentors (soph/junior/senior) to freshmen students.
 - d. Thoughts on admissions and admission criteria
 - e. Discussion on FESP program and recommendations to revising ENGR100 were discussed; issue regarding department buy-in to make ENGR100 more design-oriented, with need for engineering faculty resources (time).
4. Next meeting date TBD (late September)

Respectfully submitted,

Alan Zdunek, Secretary

Minutes of the Tuesday October 20, 2015 College of Engineering Educational Policy Committee Meeting Location 115 SEO

In Attendance: Vahe Caliskan, Joe Hummel, John Hetling, Eduard Karpov, Ludwig Nitsche, Alan Zdunek, Carmen Lilley

1. **Meeting minutes** from September meeting approved with one minor edit
2. **Admission acceptance criteria**; ACT>25, GPA>2.5, not a good correlation (13-20%) OIR looking at implementing a Selection Index (SI) that covers cases of high HS GPA and low ACT, or lower HS GPA and high ACT by estimating a first year college GPA. (handout, showing SI of 3.0 (red/yellow) and 2.5 (orange). This potentially could add 300 students to be considered for acceptance, but may introduce additional needs for these students that would have to be addressed. No action on this item needed by EPC at present time.
3. **University-wide educational assessment**: Discussed whether the engineering depts would be interested in using TK-20 program for assessment and get a demonstration. EPC agreed not to pursue this.
4. **CS109/107/111 Discussion**: Discussed whether all three are needed
 - a. ECE uses 107(C language, no Matlab), CS uses 111 (Java), other engineering use 109 (C++ and Matlab).
 - b. How to handle transfers that have taken a course in programming, but doesn't transfer 'correctly'. (Example: If ECE transfer has a programming course do they still need to take 107? ECE may need to come up with a department recommendation).
 - c. The role of a computer course in the engineering curriculum was discussed Is Matlab needed? MIE/BioE/ChE/ECE say yes (but CE part of ECE wants more programming), CME OK with it.
 - d. Committee decided that 109 stays and since 107/111 have similar objectives, CE and CS should discuss which course fits their needs; Joe Hummel will get feedback from instructors teaching courses with 107 or 111 as pre-req.
5. **CME 419 Structural Loads Determination**: Approved: Minor edit in Major topics and add more description in Course Description of types of loads.
6. **Revisions to BS CME program**: Remove CME 409, Add CME 413, CME 414, Number and title change: CME 216 to CME 322 Environmental Engineering. Approved, Minor edit to add bullet items in justification.
7. **LAS Mathematics changes**: How can COE address 2 credit hour decrease? Should COE introduce their own math course? Based on Ludwig communication with ABET, LCN and the EPC committee generally cannot recommend or support developing math courses by the engineering departments for ABET purposes. MIE and CS will review existing courses, CME will use linear algebra Math 310. CHE, BIOE, ECE to revisit.
8. **UIC TAG program**: Handout to be reviewed by EPC.
 - a. What is the most engineering courses that can be substituted for gen eds at community colleges?
9. **International minor in Engineering**: New minor being discussed in LAS and MEERP. EPC recommends focusing on getting international company interest first, not putting together the minor and seeing if there is an interest.
10. **ENGR100 revision**: LCN proposed a possible curriculum; needs further discussion.

Respectfully submitted,
Alan Zdunek,
Secretary, EPC

Minutes of the Tuesday November 2, 2015

College of Engineering Educational Policy Committee Meeting

Time: 2pm - 4pm

Location 115 SEO

In Attendance: Vahe Caliskan, Houshang Darabi, John Hetling, Eduard Karpov, Miiri Kotche, John Lillis, Ludwig Nitsche, Alan Zdunek

1. **Meeting minutes** from October 20th meeting approved
2. **ENGR100 Revision:** Suggested inclusion of an organized set of industry speakers to come and talk to students, with follow up assignments with students. Time commitment issue of speakers would be a problem. After further discussion, **EPC recommends a combined ENGR100 class with all freshman students for general engineering orientation curricula.** This should alleviate time/resource issues for student and instructors. Some specifics regarding the revision: 1) Include industry speakers for all students (schedule to be worked out for this portion of ENGR100), this could be held in the forum or another large lecture hall; 2) departments responsible for discipline specific speakers and coordinated with Chris Kuypers, 3) General technical projects (mini-projects) can also be performed (but need a large room, 250 SES was suggested), 4) Also need to have several weeks for department specific projects with a department room to accommodate students in each department. FESP committee will discuss this revision and give feedback to EPC about how to proceed.
3. TAG agreements: **LCN will resend documents, EPC will review with comments by next week (11-9).**
4. CS 202 drop: CS department recommends ECE program change to reflect CS 202 drop. **CS and ECE will discuss offline to work out details.**
5. CME 290 change: **EPC approved.** Reduce credit hours 3 to 2.
6. BioE changes: **EPC approved revisions:** 423 (new course) Approved, Pre-req changes for 325, 421, 439 (with minor change), 455, 460, 480 (minor change) pre req changes
7. Math changes to pre-calculus sequence- LCN to resend letter to EPC. Discussion about adding more relevance to MATH 180/181 through engineering examples. LCN to work out some categories. **EPC will verify that Math changes have been approved by their department.**
8. MIE and CS proposed courses to count as mathematics course for ABET: IE 342 Prob & Stats were suggested.

Respectfully submitted,

Alan Zdunek, EPC Secretary

Minutes of the Thursday December 3rd, 2015

College of Engineering Educational Policy Committee Meeting

Time: 12- 2pm

Location 115 SEO

In Attendance: Vahe Caliskan, Houshang Darabi, John Hetling, Eduard Karpov, Miiri Kotche, John Lillis, Ludwig Nitsche, Alan Zdunek

1. Meeting minutes from November 20th meeting Approved

2. ENGR100 Revisions:

Agreed on a curriculum to revise ENGR100 and to be reviewed by FESP committee; keep the existing ENGR100 curriculum (Chris Kuypers part) and the FESP-type projects and include industry speaker seminars and department information sessions; Revision will require a large lecture hall for all freshman students (1st 6 weeks), and smaller rooms for each of the department breakout sessions (last 9 weeks) and require a change in the course schedule to find an appropriate time that does not conflict with other freshman courses.

More specifically:

(1) 6 weeks of general sessions with all freshman attending for the industry related seminars (**Departments responsible for one speaker: BioE, ChE, CME, ECE/CS, and MIE (2 speakers))**).

(2) 3 weeks of department related information (**organized by the Dept DUGS**)

(3) 6 weeks Department specific/FESP technical project (**Organized by FESP**)

3. CME Course Changes: In response to MATH 180/181 credit hour decrease and ABET math/science requirements; add MATH 310, STAT 381. **EPC approved, with change in Description (delete Course Schedule change)**, Change CME396 from 2 to 2-3 credit hours and change topic list. Change CME397 to CME497 (2-3 credit hours). **EPC approved with appropriate changes to Course schedule to recognize two 'tracks' of students.**

4. ROTC programs: ROTC request that MILS courses are eligible as a free elective for COE students. They also would like COE support that MILS 217 and two other 300 MILS courses become free electives. EPC approves that MILS 217 can become a Gen Ed course (free elective).